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Predicting and Explaining Traffic Collision Severity in Toronto Using Explainable Machine Learning

Abstract

Traffic collisions remain a significant public safety concern in Toronto, resulting in injuries, fatalities, and economic losses. Although historical reports summarize collision data, there is limited use of predictive and explainable analytics to assess severity risk proactively.

This research proposes a machine learning–based framework to predict and explain traffic collision severity using publicly available City of Toronto Open Data. Multiple classification models—including Logistic Regression, Random Forest, Gradient Boosting, and XGBoost—will be developed to estimate the likelihood of minor, major, or fatal outcomes based on temporal, environmental, and spatial features.

The study will incorporate Explainable Artificial Intelligence (XAI), specifically SHAP, to interpret key risk factors and apply spatial analysis to identify high-risk areas. By combining predictive modeling with interpretability and geospatial insights, this research aims to support data-driven transportation planning and improve urban road safety in Toronto.

Problem Definition

Traffic collisions in Toronto continue to result in significant injuries, fatalities, and economic costs each year. While the City of Toronto provides detailed collision data, most analyses remain descriptive and do not offer predictive insights into collision severity.

Collision outcomes are influenced by multiple interacting factors such as time, location, weather, and road conditions. Traditional statistical summaries fail to capture these complex relationships, and predictive models often lack interpretability.

There is a need for a data-driven and explainable machine learning framework that can accurately predict collision severity and identify key risk factors to support evidence-based urban safety planning.

Research Objectives

1. Analyze historical traffic collision data from the City of Toronto Open Data portal
 - Temporal factors (time of day, day of week, season)
 - Environmental conditions (weather, lighting, road surface)
 - Location-based features (intersection type, neighborhood, coordinates)
 - Vehicle and pedestrian involvement
2. Identify key predictors influencing traffic collision severity outcomes
3. Develop classification models to predict:
 - Probability of minor injury
 - Probability of major injury
 - Probability of fatal collision
4. Compare multiple modeling approaches:
 - Logistic Regression
 - Random Forest
 - XGBoost
 - Gradient Boosting Machines
 - Neural Networks
5. Incorporate explainability techniques:
 - SHAP values
 - Feature importance analysis
6. Evaluate model performance across geographic and temporal subgroups

Research Questions

1. How accurately can machine learning models predict traffic collision severity in Toronto?
2. Which temporal, environmental, and location-based factors most strongly influence severe collision outcomes?
3. Do ensemble methods (Random Forest, XGBoost, Gradient Boosting) outperform traditional statistical models such as Logistic Regression?
4. Can explainable AI techniques provide clear and actionable insights into the key drivers of severe collisions?
5. Are certain neighborhoods, time periods, or environmental conditions associated with higher predicted severity risk?
6. Does model performance vary across geographic regions or seasonal conditions within Toronto?

Dataset Overview

Primary Dataset

City of Toronto Traffic Collision Open Data

Source: City of Toronto Open Data Portal

 <https://open.toronto.ca/catalogue/>

The primary dataset consists of officially reported traffic collision records within the City of Toronto. Each record represents a single collision event and includes structured attributes related to temporal, geographic, and incident-specific characteristics.

Dataset Contents

The collision dataset includes:

- Date and hour of occurrence (OCC_DATE, OCC_HOUR)
- Month, year, and day of week (OCC_MONTH, OCC_YEAR, OCC_DOW)
- Police division and neighbourhood identifiers
- Geographic coordinates (LAT_WGS84, LONG_WGS84)
- Collision severity indicators (FATALITIES, INJURY_COLLISIONS, PD_COLLISIONS)
- Road user involvement (automobile, motorcycle, bicycle, pedestrian, passenger)

These variables provide labeled severity outcomes and structured predictors suitable for supervised machine learning models.

Secondary Dataset (Environmental Data Integration)

Government of Canada Historical Climate Data

Source: Environment and Climate Change Canada

 <https://climate.weather.gc.ca/>

To incorporate environmental context, hourly weather data for Toronto (e.g., Toronto Pearson International Airport station) will be integrated with the collision dataset. Weather variables such as temperature, precipitation, snowfall, wind speed, and visibility will be merged using collision date and hour.

Relevance to the Project

The combined dataset is well suited for this research because it provides:

- Labeled severity outcomes for classification modeling
- Temporal and spatial features for predictive analysis
- Environmental variables to evaluate weather-related risk factors
- Publicly available and anonymized data compliant with ethical research standards

The integration of collision and climate datasets enables a comprehensive and explainable analysis of

factors influencing traffic collision severity in Toronto.

Methodology

Phase 1: Data Preparation

- Download traffic collision data from the City of Toronto Open Data portal.
- Download hourly weather data from Government of Canada Historical Climate Data.
- Merge datasets using collision date and hour.
- Handle missing values and inconsistent records.
- Encode categorical variables (e.g., neighbourhood, division).
- Perform feature engineering:
 - Hour category (peak vs off-peak)
 - Seasonal indicators
 - Weather condition indicators
- Split data into training, validation, and test sets.

Phase 2: Exploratory Data Analysis (EDA)

- Analyze distribution of collision severity classes.
- Examine temporal patterns (hour, weekday, season).
- Generate spatial heatmaps of collision density.
- Conduct correlation analysis among predictors.

Phase 3: Predictive Modeling

Models to be evaluated:

- Logistic Regression (baseline model)
- Random Forest Classifier
- XGBoost Classifier
- Gradient Boosting Machines
- Multi-layer Perceptron (Neural Network)

Class imbalance handling:

- SMOTE (Synthetic Minority Oversampling Technique)

- Class-weighted loss functions

Phase 4: Explainability and Model Interpretation

- Apply SHAP (SHapley Additive Explanations) for global and local interpretation.
- Perform feature importance analysis to identify key risk factors.
- Conduct spatial clustering (e.g., DBSCAN or KMeans) to identify high-risk zones.

Phase 5: Model Evaluation

Performance metrics:

- Accuracy
- Precision
- Recall
- F1-score
- ROC-AUC
- Confusion Matrix

Additionally, subgroup analysis will be performed across:

- Different neighbourhoods
- Seasonal variations
- Time-of-day categories

Tools & Technologies

- **Programming Language:** Python
- **Data Processing & Analysis:**
 - Pandas
 - NumPy
- **Machine Learning Libraries:**
 - Scikit-learn
 - XGBoost
 - Gradient Boosting
 - TensorFlow / PyTorch (for Neural Networks)
- **Explainable AI:**
 - SHAP (SHapley Additive Explanations)
- **Geospatial Analysis:**
 - GeoPandas
 - Folium
- **Visualization:**
 - Matplotlib

- Seaborn
- **Development Environment:**
 - Jupyter Notebook
 - VS Code
- **Version Control:**
 - GitHub

Expected Outcomes

- A machine learning model that can predict whether a traffic collision is likely to result in minor injury, major injury, or fatality.
- Clear identification of the most important factors that increase the risk of severe collisions (such as time of day, location, or weather conditions).
- Easy-to-understand explanations of how the model makes its predictions using explainable AI techniques.
- A map showing high-risk areas in Toronto where severe collisions occur more frequently.
- A complete and organized research workflow that can be reused or improved in future studies.

Conclusion

This research proposes a data-driven and explainable machine learning framework to predict traffic collision severity in Toronto. By combining collision data with environmental and spatial analysis, the study aims to identify the key factors that contribute to severe outcomes.

The integration of predictive modeling and explainable AI techniques will ensure that the results are not only accurate but also interpretable and useful for decision-makers. The findings are expected to support evidence-based transportation planning and contribute to improving urban road safety in Toronto.